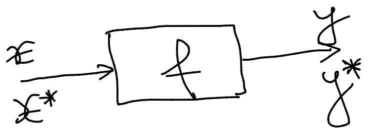
Nr. de cond. a unei probleme



$|\Delta x| = |x - x^*| \to$ er. abs. la input

$|\Delta y| = |y - y^*| \to$ —"— output

$|\delta x| = \dfrac{|\Delta x|}{|x|} \to$ er. rel. la input

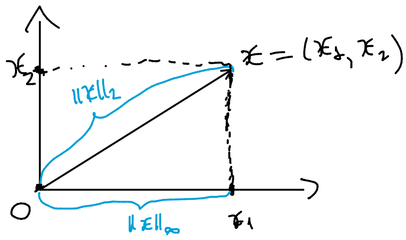
$|\delta y| = \dfrac{|\Delta y|}{|y|} \to$ er. rel. la output

$$|\delta y| \le \text{cond } f(x) \cdot |\delta x|$$

$f: \mathbb{R}^m \to \mathbb{R}^n$

$x = [x_1, \dots, x_m]^T$

$m=2$



Ex. de norme

- norma Euclidiană: $\|x\|_2 = \sqrt{x_1^2 + \dots + x_m^2}$
- norma Cebîșev/supremum: $\|x\|_\infty = \max\{|x_1|, \dots, |x_m|\}$
- norma taxi/Manhattan: $\|x\|_1 = |x_1| + \dots + |x_m|$

$A \in \mathcal{M}_{n,m}(\mathbb{R})$, $\|A\|_p = \max\limits_{\substack{x \in \mathbb{R}^m \\ x \neq 0_m}} \dfrac{\|Ax\|_p}{\|x\|_p} \to$ normă matricială

2 def. pt. nr. de cond. pt. $f = (f_1, \dots, f_n)$

$x = (x_1, \dots, x_m)$

1) $\Gamma(x) = \left(\dfrac{x_j \cdot \frac{\partial f_i}{\partial x_j}(x)}{f_i(x)} \right)_{\substack{i=\overline{1,n} \\ j=\overline{1,m}}}$ $\qquad \text{cond}_p f(x) = \|\Gamma(x)\|_p$

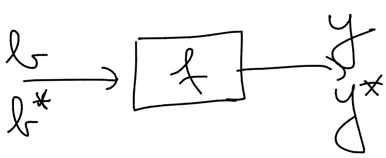
2) $\dfrac{\partial f}{\partial x}(x) := \left(\dfrac{\partial f_i}{\partial x_j}(x) \right)_{\substack{i=\overline{1,n} \\ j=\overline{1,m}}}$ $\qquad \text{cond}_p f(x) = \dfrac{\|x\|_p \cdot \left\|\frac{\partial f}{\partial x}(x)\right\|_p}{\|f(x)\|_p}.$

$f: I \to \mathbb{R}$, $f \in C^1(I)$, $I \subseteq \mathbb{R}$, $x \in I$, $f(x) \neq 0$ $\Rightarrow$ $\text{cond } f(x) = \left| \dfrac{x \cdot f'(x)}{f(x)} \right|.$

Ex. sist. lin.: $(A \in \mathcal{M}_{m,m}(\mathbb{R}))$

$A \cdot y = b$ $(\det A \neq 0)$ $\Rightarrow \exists!\, y$ sol. sist.

$y = A^{-1} \cdot b$



$f: \mathbb{R}^m \to \mathbb{R}^m$

$f(b) = A^{-1} b$

$\|b\| \cdot \|A^{-1}\|$

$b^* \quad \boxed{} \quad y^* \qquad\qquad T \quad f(b) = A^{-1} b$

$$\text{Def. 2)} \Rightarrow \quad \text{cond}_p f(b) = \frac{\|b\|_p \cdot \|A^{-1}\|_p}{\|A^{-1} b\|_p}$$

$$\text{cond}_p A = \max_{b \neq 0_n} \text{cond}_p f(b) = \dots = \|A\|_p \cdot \|A^{-1}\|_p .$$

(comandă Octave)

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>> cond(A,p)
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Ec. pol. $\quad p(x) = x^n + a_1 x^{n-1} + \dots + a_{n-1} \cdot x + a_n = 0$

$\xi \rightarrow$ răd. simplă $(p'(\xi) \neq 0,\ p(\xi) = 0)$
nenulă

$a = (a_1, \dots, a_n)$

$$\underset{a^*}{\xrightarrow{a=(a_1,\dots,a_n)}} \boxed{f} \longrightarrow \underset{\xi^*}{\xi(a)}$$

$\xi = \xi(a)$

$$\text{Def. 1)} \Rightarrow \quad \text{cond}_1 \xi(a) = \frac{\sum_{j=1}^{n} |a_j \cdot \xi(a)^{n-j}|}{|\xi(a) \cdot p'(\xi(a))|} .$$